GFN1000
(2024-25 Semester 2)

In Dialogue with Nature
与自然对话

Lecturer
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Office hour is flexible but please make an appointment in advance via email.

Course Description
Humans have long been curious about Nature. In the West, ancient Greek philosophers took the lead in exploring the physical world and the world of life with reason and hence laid the foundations of modern science. Scientists discovered physical laws, the laws governing life and even ways of controlling life. Implications of these discoveries force the humankind to re-assess their understanding of human understanding. On the oriental side of the globe, Chinese philosophers developed a different view of Nature. The comparison between Western and Chinese views will contribute to our understanding of both.

This course invites students to explore how the humankind investigated, understood and changed Nature, to study and compare the development of science in Western and Chinese cultures, and to reflect on the hu-mans' place in Nature. Students will be required to read, discuss and write about a wide range of texts in philosophy, science and its history. Reading materials will be selected from influential literatures. Emphasis will be placed on students' capacity to respond critically to these texts in written as well as oral presentations.

Textbook
In Dialogue with Nature

Course Content and Core Readings
The course is composed of 3 parts, each of which consists of several issues. Students have to read the core readings related to the issue before attending the discussion session. All texts can be found in the textbook.

Part I. Human Exploration of the Physical Universe

1. What is real?

- Plato, *Republic*/translated by C.D.C. Reeve (Indianapolis: Hackett Publishing, 2004). [Book VII (Verses 514-517)]
- David C. Lindberg, *The Beginnings of Western Science*, 2nd edition (Chicago: University of Chicago Press, 2007). [Chapter 2 (Para. 28-39)]

2. How do we handle what Nature tells us?

- David C. Lindberg, *The Beginnings of Western Science*, 2nd edition (Chicago: University of Chicago Press, 2007). [Chapter 3 (Para. 1-41), Chapter 12 (Para. 1-2, 33- 55)]

3. What is scientific understanding?

- I. Bernard Cohen, *The Birth of a New Physics* (New York: W. W. Norton & Company, 1985). [Chapter 7 (Para. 1-25, 62-63)]
- Isaac Newton, *The Principia*/A new translation by I. Bernard Cohen and Anne Whit-man. (Berkeley: University of California Press, 1999). ["Definitions" 1-5, Para. 2 of p. 408, and "Axioms, or the Laws of Motion" Corollary 1]

Part II. Human Exploration of the World of Life

4. What are the laws of life?

- Charles Darwin, *The Origin of Species*, 1st Edition. (Full text available online: http://darwin-online.org.uk/) [Chapter 4 (Para. 1-6, 9-18, 39-46, 50-63, 68-71)]

5. What is the code of life?

- James D. Watson with Andrew Berry, *DNA: The Secret of Life* (New York: Alfred A. Knopf, 2003). [Chapter 1 (Para. 1-36), Chapter 2 (whole)]

6. What impacts does manipulation of life bring?

- Rachel Carson, *Silent Spring* (Boston: Houghton Mifflin, 1962). [Chapter 6]

Part III. Our Understanding of Human Understanding

7. What guides the scientist's mind?

- Henri Poincaré, *The Value of Science: Essential Writings of Henri Poincaré* (New York: Modern Library, 2001). [*Science and Method*, Chapters I and III]

8. What is the human mind?

- Eric R. Kandel, *In Search of Memory: The Emergence of a New Science of Mind*. (New York: W. W. Norton & Company, 2006). [Chapter 4 (Para. 1-9), Chapter 28 (whole)]

9. What did Chinese know about Nature?

- Joseph Needham, *The Shorter Science and Civilization in China* Vol. 1 (Cam-bridge: Cambridge University Press, 1978). [Chapter 10 (Para. 1-3, 13-42)]

10. What has the scientific revolution revolutionized?

- Nathan Sivin, 'Why the Scientific Revolution Did Not Take Place in China – or Didn't it?' Web version: (revised 2005.8.24) http://ccat.sas.upenn.edu/~nsivin/sci-rev.pdf
- 沈括（著），胡道靜（校注），《新校正夢溪筆談》（香港：中華書局，1975）。(English translation available in the textbook) [304 節「棋局都數」，307 節「活版印刷」，357節「虹」，430 節「海陸變遷」，437 節「指南針」]

11. What makes modern science unique?

- William Dunham, *The Mathematical Universe: An Alphabetical Journey Through the Great Proofs, Problems, and Personalities* (New York: Wiley & Sons, 1994). [Chapter G]
- Euclid, *Elements* / translated by Thomas L. Heath. Web version: http://www.per-seus.tufts.edu [Book 1: "Definitions", "Postulates", "Common Notions", and "Propositions" 1-5, 7-11, 13, 15-16, 18-20.]

Course Schedule

Lecture	GFN1000	Fri 13:30-14:20, TD307
Tutorials	GFN1000-T20	Mon 15:30-17:20, TC 114
	GFN1000-T19	Tue 15:30-17:20, TC 302
	GFN1000-T21	Tue 13:30-15:20, TC 302

Week	Dates (Mon-Fri)	Tutorial (Mon &Tue)	Lecture (Fri)	Remarks
1	*Jan 6 - 10*	Course Introduction	01 Plato & Lindberg-1	
2	*Jan 13 - 17*	01 Plato & Lindberg-1	02 Aristotle & Lindberg-2	
3	*Jan 20 – Feb 7*	**Chinese New Year Holiday**		
4	*Feb 10 - 14*	02 Aristotle & Lindberg-2 **Quiz 1**	03 Cohen & Newton RJ Topics	
5	*Feb 17 - 21*	03 Cohen & Newton	04 Darwin	
6	*Feb 24 - 28*	04 Darwin	05 Watson	Mind map I submission
7	*Mar 3 - 7*	05 Watson **Quiz 2**	06 Carson	**RJ submission**
8	*Mar 10 - 14*	06 Carson	07 Poincaré	
9	*Mar 17 - 21*	07 Poincaré	08 Kandel	
10	*Mar 24 - 28*	08 Kandel **Quiz 3**	09 Needham	
11	*Mar 31 – Apr 4*	**Holiday**		
12	*Apr 7 - 11*	09 Needham	10 Sivin &Shen TP Topics	
13	*Apr 14 - 18*	10 Sivin & Shen **Quiz 4**	11 Dunham& Euclid	
14	*Apr 21 – 25*	11 Dunham&Euclid	TP consultation	
15	*Apr 28 – May 2*	TP consultation		Mind map II submission
16	*May 5 - 9*			**TP ddl May 11, 2025**

Note:
1. *The schedule is tentative and may be subject to change.*
2. Details of mind map will be announced later.
3. Deadline of reflective journal submission: Mar 9 (Sun, 23:55), 2025.
4. Deadline of term paper submission: May 11 (Sun, 23:55), 2025.

Assessment

1. Participation and Engagement (30%):

- **In-class Discussion (25%)**
  Students are required to participate actively in class discussion. Marks will be given with reference to students' attendance and performance in the lectures and tutorials on understanding the texts, responding to questions, presenting (individually) ideas and discussion results, and playing the role of novice leader.

- **Mind maps (5%)**
  Students are tasked with creating two mind maps that reflect the content covered in our course.

2. Quizzes (25%):

- Four quizzes (in English) will be conducted in tutorials. They are mainly in multiple-choice form and based on the content of the textbook.

3. Reflective Journal (RJ) (15%):

- Students are required to finish a reflective journal at mid-term: 600-800 words in English or 900-1,200 characters in Chinese. Deadline of reflective journal submission: Mar 9 (Sun, 23:55), 2025.

4. Term Paper (TP) (30%):

- Students are required to finish a term paper at the end of the course: 1,300–1,500 words in English or 1,900–2,300 characters in Chinese. Deadline of term paper submission: May 11 (Sun, 23:55), 2025.

Rules and Absence Policy

- Quizzes will start at the beginning of classes. **No make-up quiz** will be available for late comers.
- **Late submission** of assignments may incur **penalty** or may be granted zero mark.
- You are allowed **only one non-excused absence**. For excused absence, you have to seek your teacher's prior approval, or in the case of illness, bring in a medical certificate afterwards. Too many absences (excused or not) may affect your final marks or result in your failure of the course.

Academic Honesty and Avoid Plagiarism

- AI usage is prohibited in this course. For detailed regulations, please refer to the 'Updated_Use_of_VeriGuideAWD_August2024' document.
- Attention is drawn to University policy and regulations on honesty in academic work, and to the disciplinary guidelines and procedures applicable to breaches of such policy and regulations. Details may be found at http://www.cuhk.edu.hk/policy/academichonesty/.
- With each assignment, students will be required to submit a signed declaration that they are aware of these policies, regulations, guidelines and procedures. For group projects, all students of the same group should be asked to sign on the declaration.
- For assignments in the form of a computer-generated document that is principally text-based and submitted via VeriGuide, the statement, in the form of a receipt, will be issued by the system upon students' uploading of the soft copy of the assignment. Assignments without the receipt will not be graded by teachers. Only the final version of the assignment should be sub-mitted via VeriGuide.

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